

Hybrid Keyword and Semantic Retrieval

Student Name:	Student Name	Student ID / Roll:	EXP-013
Experiment Date:	2026-09-18	Quiz Evaluation:	0 / 10 (0%)

1. Learning Objectives

- Understand how BM25 scores documents using term frequency and inverse document frequency.
- Understand how embedding-based semantic search captures meaning beyond exact keyword overlap.
- Combine lexical (BM25) and semantic scores into a single hybrid ranking using a tunable weight alpha.
- Analyze how alpha, k1 and b change document rankings for different queries.
- Evaluate whether hybrid retrieval improves result relevance compared to either method alone.

2. Recorded Experimental Trials

No simulation trials recorded during this session.

3. Observations & Analysis

Hybrid retrieval was compared against BM25-only and semantic-only rankings across several queries, showing improved coverage of semantically related but lexically different documents.

Instructor / Student Signature